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MILITARY SPECIFICATION

CUTOUT RELAY, ENGINE GENERATOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of 28-volt direct current (DC) cutout relay to be furnished with the ratings shown on MS27443 as specified.

2. APPLICABLE DOCUMENTS

*2.1 Government documents.

*2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Shipping, Fiberboard

MILITARY

DOD-D-1000	Drawings, Engineering And Associated Lists
MIL-M-14	Molding Plastics And Molded Plastic Parts Thermosetting
MIL-P-116	Preservation, Method Of
MIL-B-83769	Battery, Storage, Lead Acid, General Specification For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-S-19500	Semiconductor Device, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-704	Aircraft, Electric Power, Characteristics
MIL-STD-794	Parts And Equipment Procedure For Packaging Of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS18045	Batteries, Storage, Aircraft, 23-Volt General Purpose
MS27433	Cutout Relay, Engine Generator 18-Volt DC System

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 3951 Standard Practice For Commercial Packaging

(Application for copies should be addressed to Amercian Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules And Regulations

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago IL 60606.)

*2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

*3.1 First Article testing. This specification makes provision for First Article testing.

3.2 Materials. Materials used in the manufacture of units shall be suitable for the purpose, and shall conform to applicalbe Government specifications. Materials conforming to contractor's specification may be used, provided the specifications are approved by the Government and contain provisions for adequate tests. The use of the contractor's specifications will not constitute waiver of Government inspection.

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3.2.1 Materials shall be of a corrosion-resistant type or suitably processed to resist corrosion during the normal service life of the cutout.

*3.2.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.3 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.2.3.1 Standard parts. Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part number are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction. The cutout shall be designed to automatically connect and disconnect a generator from the bus in a 28-volt DC system as defined in MIL-STD-704. In connecting the generator to the bus, the cutout shall close when the generator is at 18 +6 -0 volts and is 0.5 $\pm$0.15 volt above bus potential. The cutout shall close only when positive potential is applied to the "GEN" terminal. When the generator voltage decreases below bus voltage, the cutout shall open with a reverse current within the limits specified on the applicable drawing. In addition, the cut-out shall be so designed as to make it impossible to transfer to the "SW" terminal a positive potential that is impressed upon the "APP" terminal after generator shut-down with no reverse current.

3.3.1 Assembly. The cutout assembly shall be mounted on a base which is designed for permanent installation on aircraft. The unit shall not be of a plug-in type. The main terminal base of the cutouts shall provide good electrical insulating properties, impact strength, and shall resist moisture absorption and cracking or warping that is due to temperature extremes encountered. Cellulose-filled plastic materials shall not be acceptable. The component parts of the cutouts shall be so designed and attached to the supporting members that stress up during mounting of the cutout shall not adversely affect its operation.

3.3.2 Creepage and clearance distances. The minimum creepage path between current-carrying parts and ground shall be 1/8 inch. The minimum clearance between current-carrying parts and any other portion of the cutout other than insulating material, and also between current-carrying parts of opposite polarity, shall be 1/16 inch.

3.3.3 Terminal studs. All terminal studs on the cutout, to which external connections are made, shall be designed to prevent rotation of the stud when installing and removing terminals.

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3.3.4 Adjustment. All adjustments for satisfactory operation of the cutout shall be made by the manufacturer before delivery to the procuring activity. The cutout shall operate satisfactorily in service for 1,000 hours without further adjustments.

3.3.5 Case insulation. The case, if metal, shall be lined with heat-resistant insulating coating which will not crack or craze when subjected to -54° Centigrade (C) and will not fail when subjected to copper heated to 982°C.

3.3.5.1 The molded insulated top of the relay shall be composed of type MA1-60, mineral filler alkyd resin or glass mat polyester in accordance with MIL-M-14.

3.3.6 Semiconductor devices. Semiconductor devices used in the cutout shall meet the requirements of MIL-S-19500 and shall be tested in accordance with 4.6.4.3.

*3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.5 Terminal screw threads. Terminal screw threads used externally shall conform to the requirements of MIL-S-7742.

3.6 Weight and dimensions. The maximum weight and over-all dimensions of the cutout shall be in accordance with the values shown on MS27433.

3.7 Performance. The cutouts shall meet the requirements of this specification when subjected to the following tests:

(a)	Examination of product	4.6.1
(c)	Potential drop	4.6.3
(d)	Overload	4.6.4
(e)	Surge load	4.6.5
(f)	Interrupting capacity	4.6.6
(g)	Endurance	4.6.7
(h)	Inductive load	4.6.8
(i)	Reverse current due to loss of generator field	4.6.9
(j)	Reverse voltage	4.6.10
(k)	Insulation	4.6.11
(l)	Case insulation	4.6.12
(m)	Main contactor closing and opening speed	4.6.13
(n)	Cutout operating time	4.6.14
(o)	Cold temperature	4.6.15
(p)	High temperature	4.6.16
(q)	Shock	4.6.17
(r)	Acceleration	4.6.18
(s)	Vibration	4.6.19
(t)	Sand and dust	4.6.20
(u)	Humidity	4.6.21
(v)	Fungus resistance	4.6.22
(w)	Salt spray	4.6.23
(x)	Explosion-proof (Aeronautical)	4.6.24

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3.8 Identification of product. The cutout shall be marked in accordance with MIL-STD-130, to provide the following information, in the order shown:

CUTOUT RELAY, AIRCRAFT GENERATOR
28-Volt DC _____ Amperes
MS27433 (proper dash number)
Stock No.
Manufacturer's Part No. (or identification)
Manufacturer's Serial No.
Manufacturer's Name or Trade-Mark

*3.8.1 Use of AN or MIL designations. AN or MIL designations shall not be applied to a product, except for first article samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

3.9 Workmanship. All machined surfaces shall have a smooth finish, and all details of manufacture, including the preparation of parts and accessories shall be in accordance with the best practice for high-quality aircraft cutouts. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts, plating, lacquering, riveting, clearance between soldered connections, and ruggedness.

3.10 Humidity. Relays tested as specified in 4.6.21 shall show no evidence of corrosion, breaking, cracking, or loosening. Terminal and mounting hardware shall be readily removable.

3.11 Fungus resistance. Relays tested as specified in 4.6.22 shall meet the requirements specified for adjustment in 4.6.2.

3.12 Salt spray. Relays tested as specified in 4.6.23 shall show no evidence of excessive corrosion. Excessive corrosion is defined as that amount of contamination which would interfere with the electrical/mechanical performance of the relay. In addition, there shall be no evidence of cracking or other visible damage to the relay.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of tests. The inspection and testing of cutouts shall be classified as follows:

a. First article tests

b. Acceptance tests

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*4.3 Test. First article test shall include all tests of 4.6 and the requirements of MS27433 in a sequence as outlined in Table I and the footnote thereto.

*4.3.1 Test sample. A maximum of four (4) first articles shall be tested for design approval. Test specimens may be used by the contractor as a first article sample or submitted for acceptance at the end of the contract provided they are completely refurbished to comply with the requirements stated herein. All parts specimens or assemblies destroyed in making tests required by this specification shall be in addition to the quantity specified in the contract or purchase order.

*4.3.2 Test report. A certified copy of the contractor's first article test data to MIL-STD-831 shall be submitted to the procuring agency and shall include an analysis of all failure and remedial action taken. A record of all tests and inspections required to determine compliance with the requirements specified herein shall be kept complete and available to the Government at all times. The test data shall include representative oscillograms of all circuit-interrupting tests and shall be signed and approved by a responsible person specifically designated by the contractor.

TABLE I - Testing sequence. 1/

Cutout No. 1		Cutout No. 2	
Examination of product	4.6.1	Examination of product	4.6.1
Adjustment	4.6.2	Adjustment	4.6.2
Potential drop	4.6.3	Potential drop	4.6.3
Insulation	4.6.11	Insulation	4.6.11
Overload	4.6.4	Endurance	4.6.7
Surge load	4.6.5	Inductive load	4.6.8
Interrupting capacity	4.6.6		

1/ The other required tests may be run on the above two cutouts or on other units, as desired. Sequence of testing of these other tests is optional, except, that Cold- and High-temperature tests (4.6.15 and 4.6.16, respectively) shall be conducted on the same unit. If fungus resistance and salt spray tests (4.6.22 and 4.6.23, respectively) are conducted on the same unit, the fungus resistance (4.6.22) shall be first. Tests on any additional units shall be preceded by tests for examination of product (4.6.1), adjustment (4.6.2), potential drop (4.6.3), and insulation (4.6.11).

4.4 Acceptance.

4.4.1 Quality conformance. Production relays shall be examined and tested in accordance with sampling plans Group A and Group B. (See 4.4.2 and 4.4.3).

4.4.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in Table II, and shall be made on each relay submitted for acceptance. There shall be zero acceptance for test failures.

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4.4.3 Group B inspection. Group B inspection shall consist of the examination and tests specified in Table III. Statistical sampling in accordance with MIL-STD-105 shall be performed using level S-3 for acceptance quality levels indicated.

4.5 Acceptance and rejection. When any sample fails to meet the requirements as set forth herein and on MS27433, resubmission of lots shall be in accordance with MIL-STD-105. In addition, full particulars concerning rejection and corrective measures shall be furnished the Government representative. Units rejected after retest shall not be resubmitted without specific approval of the procuring agency.

TABLE II - Group A inspection.

Examination or Test	Requirement Paragraph	Method Paragraph
Examination of Product	3.2	4.6.1
Adjustment	3.3 and 3.3.4	4.6.2 1/
Potential Drop		4.6.3
Insulation		4.6.11

*1/ Operation for 1,000 hours shall be demonstrated during first article testing only.

TABLE III. Group B inspection.

	Test	Test Paragraph	AQL (Percent Defective) Major	Minor
	Examination of Product	4.6.1	1.5 percent	4
	Endurance	4.6.7	1.0 percent	N/A
	Reverse Current Due to Loss of Generator Field	4.6.9	1.5 percent	N/A
	Reverse Voltage	4.6.10	1.5 percent	N/A
	Case Insulation	4.6.12	2.5 percent	N/A
	Main Contactor Closing Speed	4.6.13	1.0 percent	N/A
	Relay Operating Time	4.6.14	1.0 percent	N/A
	Cold Temperature	4.6.15	2.5 percent	N/A
	High Temperature	4.6.16	2.5 percent	N/A
*	Shock	4.6.17	1.5 percent	N/A

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TABLE III. Group B Inspection. - Continued.

Test	Test Paragraph	AQL Major	Percent Defective Minor
Acceleration	4.6.18	1.5 percent	N/A
Vibration	4.6.19	2.5 percent	N/A

4.6 Test methods.

4.6.1 Examination of product. Each cutout subassembly and part shall be examined to determine conformance with this specification and MS27433 with respect to workmanship, construction, dimensions, weight, interchangeability, identification marks, and instructions.

*4.6.2 Adjustment. The cutout shall be tested in a system consisting of two aircraft generators connected in parallel with a battery conforming to MIL-B-83769, or in an equivalent circuit, to determine performance characteristics as follows.

4.6.2.1 Differential voltage. The cutout shall close only when the generator voltage is 18 +6 -0 volts and is 0.5 ±0.15 volts above bus potential. This test shall be performed after the cutout has been opened by reverse current during the normal generator shutdown.

4.6.2.1.1 Voltage transfer. This test shall be performed after generator shutdown with no reverse current. The cutout shall be subjected to a one minute application of 28 volts DC between the "APP" terminal (Positive) and the case (Negative). The voltage when measured between the "SW" terminal and ground shall be zero volts.

4.6.2.2 Unexcited bus. On an unexcited bus the relay shall close on a load having a resistance of 100 ohms or less before the generator voltage exceeds 24 volts.

4.6.2.3 Reverse current. The voltage of the generator shall be controlled by means of a variable resistance in the generator field circuit while the bus voltage is held constant at 28 volts. Upon decreasing the voltage of the controlled generator, the reverse current flowing immediately prior to the opening of the contacts shall not exceed the value specified on MS27433 in the column headed "Reverse Current."

4.6.2.4 Remote switch. The electrical circuit of the cutout shall be such that a suitable switch connected as shown on Figure 1, will cause the main contactor to close on battery voltage when the switch is in the momentary position for the purpose of cranking auxiliary powerplants, using the auxiliary powerplant generator as a starter. When actuated from momentary position at a maximum voltage of 18 volts and hold in while the voltage is reduced to 7 volts or less. When the switch is in the continuous position, the cutout shall operate normally. The total resistance of each external control circuit shall not exceed 1.0 ohm. When the control switch is placed in the "off" position, the main contactor shall open to disconnect the generator from the line.

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4.6.3 Potential drop. The drop in potential between the input and the output terminals shall be measured at rated current. The value shall be determined from the average of 5 readings of up to 2 minutes duration per reading. No single reading shall exceed a maximum of 125 percent of the value specified on MS27433.

4.6.4 Overload.

4.6.4.1 Main contacts. The setup for this test shall be as shown on Figure 1. The cutout shall be operated, by a remote switch, at the overload specified on MS27433 for 500 cycles consisting of 1 second on and 5 seconds off. There shall be no welding of the contacts. After completion of this test, the cutout shall be subjected to the potential drop test specified in 4.6.3, and the value shall be below 110 percent of the drop specified on MS27433. The cutout shall be subjected to the test specified for adjustment (4.6.2), and shall meet the requirements specified therein.

4.6.4.2 "IND" circuit. The "IND" circuit shall be subjected to overload currents of 10, 20, 50, 100, and 150 amperes (amps). After each application of overload current, it shall be determined that no overheating or damage occurred to the flexible lead in the "IND" circuit.

4.6.4.3 Semiconductor overload. A cutout that utilizes semiconductor devices to accomplish the requirements of 3.3 shall be subjected to the overload test of 4.6.4.3.1 and 4.6.4.3.2. The contacts of the differential relay shall be closed during the performance of this test.

4.6.4.3.1 Steady-state overload. The cutout shall be subjected to a 15 minute application of 50 volts DC between the case (Negative) and the "APP"terminal (Positive). The voltage measured between the "SW" terminal and ground shall effectively be zero volts. There shall be no indication of diode breakdown or conduction in the reverse direction. This test shall be repeated after the completion of the test of 4.6.4.3.2. Then a positive 28 volts DC shall be applied between the "SW" terminal and ground and shall be measurable between the "APP" terminal and ground.

4.6.4.3.2 Transient overload. The cutout shall be subjected to transient surge and spike voltage as defined in MIL-STD-704, Category B equipment. These voltages shall be applied in the direction of both forward and reverse condition. Measurements shall be taken during the test to insure that no diode breakdown, shorting, or conduction in the reverse direction shall occur.

4.6.5 Surge load. The cutout shall be subjected to 5,000 operations at a surge load and reduced voltage as shown in Figure 2 and Figure 3, as applicable. There shall be no welding of the contacts. After completion of this test, the cutout shall be subjected to the potential drop test specified in 4.6.3, and the value shall be below 110 percent of the drop specified on MS27433. The cutout shall meet the requirements specified for adjustment (4.6.2).

4.6.6 Interrupting capacity. The cutout shall be connected to one generator in a system of several aircraft generators and aircraft batteries in parallel. The "IND" terminal of the cutout shall be connected to a 5 amp inductive load. A "fault" circuit shall be connected between the positive terminal of the generator equipped with the cutout and the common negative

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of the system. The number of generators and batteries and the fault circuit resistance shall be such that the interrupting capacity indicated on MS27433 is obtained. Ten interruptions at intervals of 30 seconds at sea level and at a pressure corresponding to an altitude of 65,000 feet shall be made. An oscillograph shall be used to indicate opening of the cutout and for measuring the interrupted current. The differential relay shall interrupt the main contactor coil current before the main contactor opens. At the conclusion of this test, the differential voltage and the reverse current calibration shall be checked and shall not exceed specified limits. The relays shall be capable of interrupting the short circuit listed on MS27433, in column headed "Interrupting Capacity."

4.6.7 Endurance.

4.6.7.1 Sea-level. The cutout shall be subjected to 5,000 operations making and breaking rated current at rated voltage. The "IND" terminal of the cutout shall be connected to a 5 amp inductive load. The test shall be conducted at a rate of 2 seconds "ON" and 4 seconds "OFF," or faster, so long as proportionate duty cycle is maintained. After completion of this test, the cutout shall be subjected to the potential drop test specified in 4.6.3, and the value shall be below 110 percent of the drop specified on MS27433.

4.6.7.2 Altitude. The preceding test shall be repeated at a pressure corresponding to an altitude of approximately 65,000 feet. After completion of this test, the cutout shall be subjected to the potential drop test specified in 4.6.3, and the value shall be below 110 percent of the drop specified on MS27433. The cutout shall then be subjected to the tests specified for adjustment (4.6.2) and insulation (4.6.11). It shall meet the requirements specified therein. There shall be no mechanical failure.

4.6.8 Inductive load. The cutout shall be subjected to 2,000 operations of making and breaking rated current at rated voltage. The time constant of the load shall be 0.006 second. The first 1,000 operations shall be at sea level and the final 1,000 at a pressure corresponding to an altitude of approximately 65,000 feet. The cutout shall then be subjected to the tests specified for adjustment (4.6.2) and insulation (4.6.11). It shall meet the requirements specified therein. There shall be no mechanical failure.

4.6.9 Reverse current due to loss of generator field. The cutout shall be connected to one generator in a system consisting of four aircraft generators and three MS18045 aircraft batteries in parallel. The current rating of the generators shall be the same as the current rating of the cutout, except that 400 amp generators shall be used for testing MS27433-600 cutouts with the system voltage adjusted to 28.5 volts. The field circuit of the generator to which the cutout is connected shall be opened, and opening of the cutout contacts shall be determined visually. Opening of the field circuit shall be repeated at 1/2-minute intervals for 1 hour. At the conclusion of this test, the differential voltage and the reverse current calibration shall be checked and shall be within specified limits.

4.6.10 Reverse voltage. With a jumper connected across GEN and SW terminals, and with negative lead connected to GEN terminal and the positive lead connected to the BAT terminal, a potential of 60 volts shall be applied to GEN and BAT terminals for a 3-hour period. It shall be determined that the main contacts do not close. Immediately after this test, the cutout

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shall be subjected to the tests specified for insulation (4.6.11) and adjustment (4.6.2). It shall meet the requirements therein. (This test may be omitted for cutout designs in which the differential coil is protected by a polarized voltage relay.)

4.6.11 Insulation. A potential of 1,000 volts rms commercial frequency shall be applied between the terminals and other exposed metal parts for a period of 60 seconds. The coil ground connections shall be disconnected for this test. There shall be no failure.

4.6.12 Case insulation. Three pieces of copper 1/2 by 1/2 by 1/2 inch shall be heated to 98^{+2}°C and immediately placed in the bottom of the case. After the copper has cooled to room temperature, and without having been moved from its original position in the cover, a high potential test of 500 volts alternating current (AC) for a period of 1 minute shall be applied between the case and the blocks. There shall be no breakdown of insulation.

The above test shall be repeated with the three cubes placed on a side of the case, with that side lying horizontal. If the case has sides which are other than flat, the curves may be contoured to fit the case, but must have the same weight as the 1/2-inch cube, and present 1/4 square inch of surface to the insulation.

4.6.13 Main contactor closing and opening time. When energized with rated voltage from the APP terminal, the main contactor closing time shall not exceed 0.045 second. The main contactor opening time shall not exceed 0.035 second.

4.6.14 Cutout operating time. When energized with rated voltage from the SW terminal, action of auxiliary relays, auxiliary mechanisms, and main contactor coil shall be sufficiently fast to close the main contactor in less than 0.060 second. The cutout shall disconnect the generator from the bus in less than 0.040 second, after instantaneous application of reverse current equal to 10 times the maximum specified value listed on MS27433 in the column headed "Reverse Current."

4.6.15 Cold temperature. The cutout shall be placed within a test chamber and cooled to and maintained at a temperature of -54° ±2°C for a period of 72 hours. During the 72-hour period, the cutout shall be cycled for a period of 30 minutes of each 24-hour interval. During the cycling period, the load current shall be 1/20 of the cutout rating consisting of 1 second on and 5 seconds off. At the end of this period and with the cutout at the above specified temperature, the cutout shall meet the adjustment tests specified for differential voltage (4.6.2.1), unexcited bus (4.6.2.2), and reverse current (4.6.2.3). No failure or malfunctioning shall occur either during or as a result of the cycling procedure. The case insulation shall not show signs of cracks or crazing due to the cold test.

4.6.16 High temperature. The cutout shall carry rated current at a coil voltage of 29 volts in an ambient temperature of 71°C for a period of 3 hours. During the last hour of this period, the cutout shall be subjected to the tests specified for adjustment (4.6.2). The remote switch pickup voltage shall not exceed 22 volts during the adjustment test. This voltage limit applies under test conditions of this paragraph only. Following this test, the cutout shall meet the requirements specified for insulation (4.6.11).

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*4.6.17 Shock. The cutout shall be shock-tested in accordance with the shock test, Procedure I, of MIL-STD-810. The cutout shall be rigidly fastened, using the cutout mounting means, to the drop carriage of the shock testing machine and subjected to single resilient impacts of 25g intensity of 6 to 9 milliseconds duration. Four shocks shall be applied in each direction along each of three mutually perpendicular body axes, two of which shall be with the main contactor coil de-energized, and two with the coil energized. The cutout shall be energized with 18 volts applied to the coil at the APP terminal for 3 hours prior to and during the time of impact. Evidence of contact chatter shall be determined by an oscillograph or a storage scope with a permanent record made of the pulse on the storage scope. The total duration of contact chatter, over the above contact resistance fluctuations, shall be less than 3 milliseconds.

*4.6.18 Acceleration. The cutout shall be subjected to an acceleration force of 10g. The time required to attain the 10g acceleration shall be not more than 10 seconds. The cutout shall be mounted in the position most likely to cause malfunctioning and shall be accelerated for a minimum of 1 minute. During this test, the main contacts shall remain open with no voltage across the coil, and shall remain closed when the coil voltage is 18 volts. A test lamp or other indicating device shall be connected across the contacts while the cutout is undergoing this test to determine the ability of the cutout contacts to remain in the proper position.

4.6.19 Vibration. The cutout shall be subjected to simple harmonic or circular motion applied to the mounting base. The maximum deflection shall be 0.06 to 0.07 inch. The frequency shall be varied from approximately 10 to 55 and back to 10 cycles per second, the period of variation being 1 to 5 minutes. The cutouts shall be mounted horizontally and vibrated continuously for 2 hours at 29 volts, full load; and then for 1 hour unenergized. The test shall then be repeated with the axis revolved 90 degrees to the original axis. At least one of these tests shall be conducted with the plane of vibration parallel to the plane or planes of contact travel. The cutout calibration shall be determined at 0, 10, 20, 30, 40, 50, and 55 cycles per second, 0.06-inch total excursion at the end of each 3-hour period. In all cases, the calibration shall fall within the requirements specified for adjustment (4.6.2). The contacts shall not close or open any time due to vibration.

4.6.20 Sand and dust. The cutout shall be subjected to the sand and dust test of MIL-STD-810. At the end of the test, the cutout shall meet the requirements specified for adjustment (4.6.2).

4.6.21 Humidity. The cutout shall be subjected to humidity tests, Procedure I, of MIL-STD-810. During the cycling procedure, a potential of 150 volts root mean square (rms) at commercial frequency shall be applied between the terminals and other exposed metal parts. The current leakage shall not be in excess of 0.1 amp at any time during this test. After the cutout has been removed from humidity, it shall be dried for approximately 6 hours in a circulating air oven at approximately 57°C. Immediately after drying, the cutout shall be subjected to and shall satisfactorily meet the test requirements for adjustment (4.6.2) and insulation (4.6.11).

4.6.22 Fungus resistance. The cutout shall be subjected to the fungus resistance test, Procedure I, of MIL-STD-810 No component part of the cutout shall exhibit fungus growth. This test need not be conducted

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provided the manufacturer certifies that none of the material used in the cutout will deteriorate or corrode under the fungus conditions of Procedure I.

4.6.23 Salt spray. The cutout shall be subjected to the salt spray test of MIL-STD-810 for 50 hours. After the cutout has been removed from salt spray, it shall be dried for approximately 6 hours in a circulating air oven at approximately 57°C. Immediately after drying the cutout shall be subjected to and satisfactorily meet the test requirements for adjustment (4.6.2). No component part of the cutout shall exhibit any signs of corrosion.

4.6.24 Explosion-proof (aeronautical). The cutout shall be subjected to and shall satisfactorily meet the explosion-proof (aeronautical) tests, Procedure I, of MIL-STD-810. All operating tests shall be conducted at rated load with a time constant of 0.006.

4.6.25 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING

5.1 Preservation and packaging. Preservation shall be Level A, C or commercial as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. The cutout relays shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Cutout relays shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation Applications. Preservatives shall not be used.

5.1.1.4 Packaging. Unless otherwise specified (see 6.2) each cutout relay shall be individually packaged in accordance with MIL-P-116, Method IA.

5.1.2 Level C. The level C preservation for cutout relays shall conform to the requirements of MIL-STD-794.

5.1.3 Commercial. The commercial/industrial preservation of cutout relays shall be in accordance with the requirements of ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, C or commercial, as specified (see 6.2).

5.2.1 Level A. The packaged cutout relays shall be packed in containers as specified in MIL-STD-794 for Level A protection.

5.2.2 Level B. The packaged cutout relays shall be packed in containers as specified in MIL-STD-794 for level B protection.

5.2.3 Level C. The packaged cutout relays shall be packed in containers as specified in MIL-STD-794 for Level C protection.

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5.2.4 Commercial. Commercial/industrial packing of the cutout relays shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Level A, B and C. In addition to any special or other identification marking required by the contract (6.2) each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial/industrial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The cutout relays covered by this specification are intended for use with 30-volt aircraft generators on 28-volt electrical systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. The MS part number of the cutout desired.

c. Invitation for bids, contracts and purchase orders shall specify the applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

*6.3 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements fo this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AV
Air Force - 99

Review activity:
Army - AV

Preparing activity:
Air Force - 82

Project No. 2925-0017

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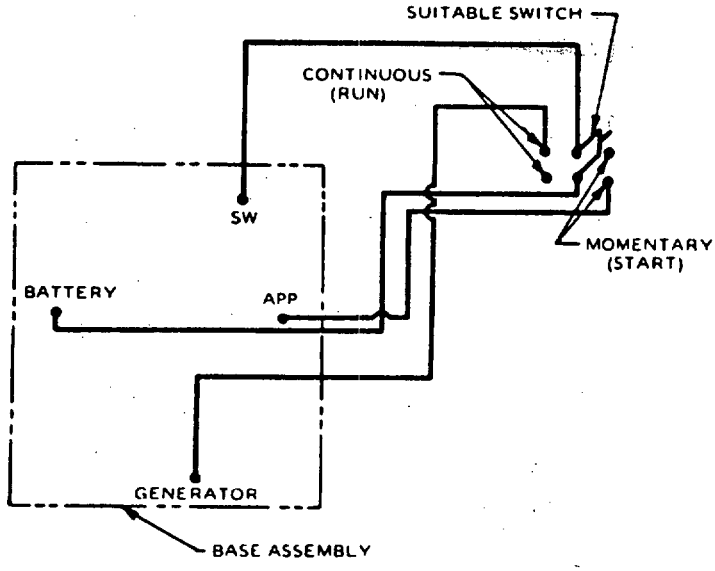


FIGURE 1. SETUP WITH REMOTE SWITCH

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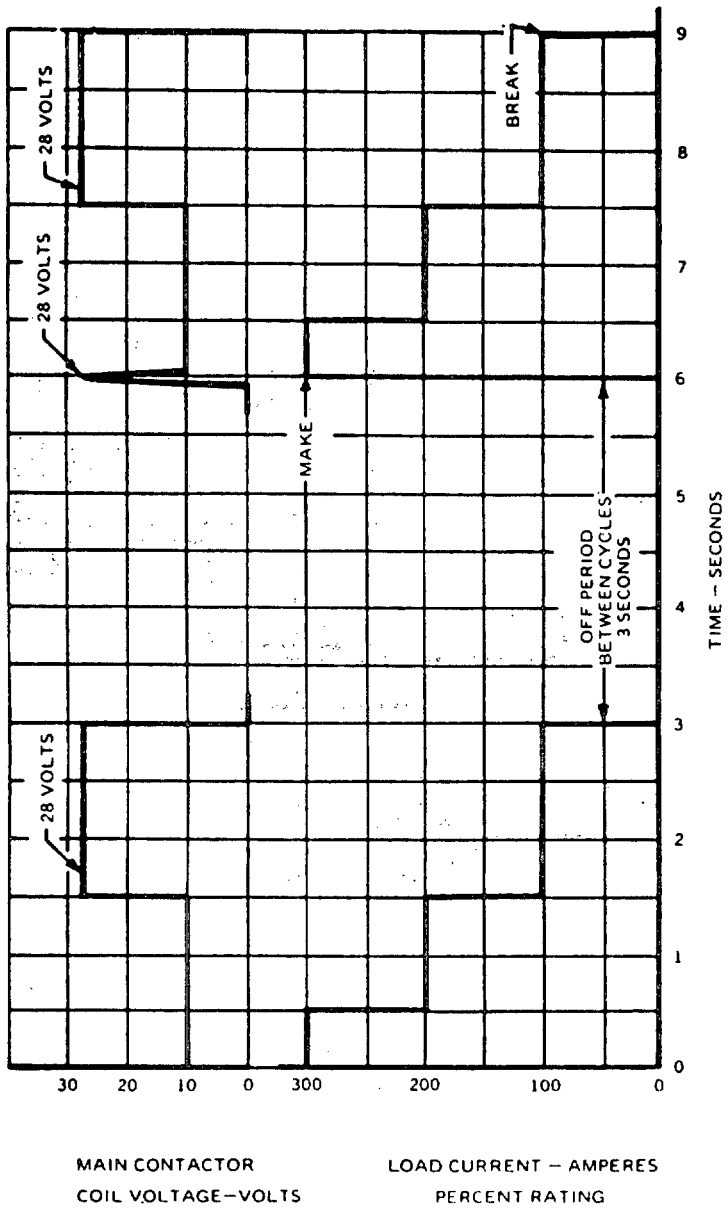


FIGURE 2

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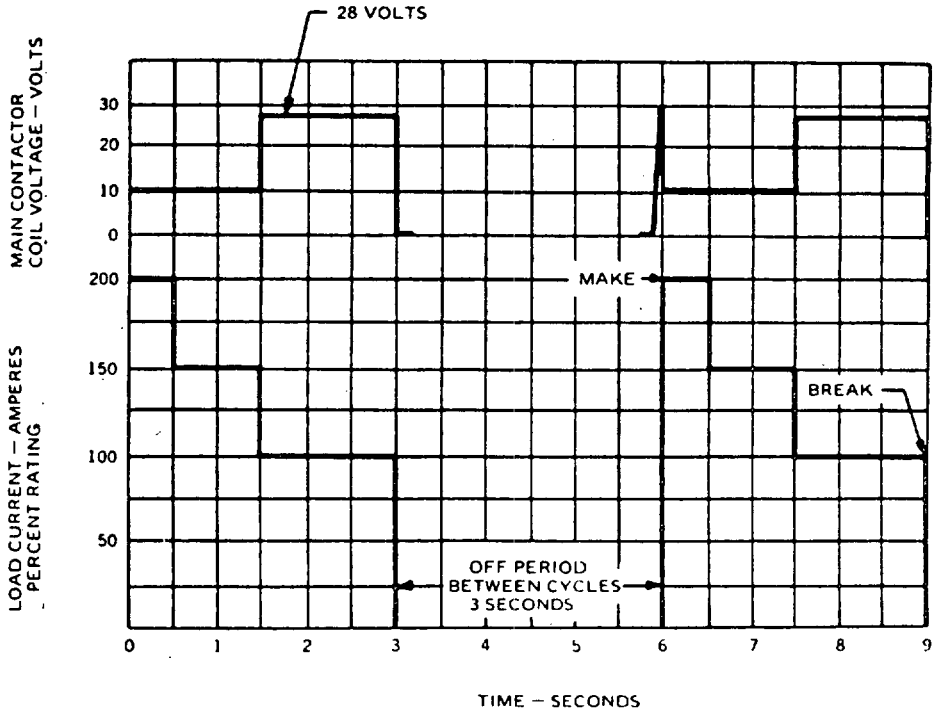


FIGURE 3

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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